Muhammad Rakha Naufal Maulana A0330820U

1. Non-relativistic Limit

Let's start from the Lorentz boost

$$x' = \gamma(x - vt),\ t' = \gamma\left(t - \frac{vx}{c^2}\right)$$

for $v \ll c$, we have $\gamma \approx 1$
and $\frac{vx}{c^2} \to 0$
Thus, we get

$$x' \approx x - vt,\ t' \approx t$$

Q.E.D.

2. Composition of Co-linear Boosts

Using Lorentz boost,

$$t' = \gamma\left(t - \frac{vx}{c^2}\right),\ x' = \gamma(x - vt)$$

differentiating the two equations gives us

$$dt' = \gamma\left(dt - \frac{vdx}{c^2}\right),\ dx' = \gamma(dx - vdt)$$

taking $\frac{dx'}{dt'}$, we get

$$\frac{dx'}{dt'} = \frac{\gamma(dx - vdt)}{\gamma\left(dt - \frac{vdx}{c^2}\right)} = \frac{\left(\frac{dx}{dt} - v\right)}{1 - \left(\frac{vdx}{dtc^2}\right)}$$

Using $V = \frac{dx}{dt}, u = \frac{dx'}{dt'}$ and rearranging gives us

$$u = \frac{V - v}{1 - \frac{Vv}{c^2}} \Rightarrow V = \frac{v + u}{1 + \frac{uv}{c^2}}$$

3. Composition of Boosts in Different Directions

a)
Let's first do the boost in the z-direction

$$z' = \gamma_u(z - ut),\ ct' = \gamma_u(ct - \beta_u z),\ x' = x, y' = y$$

now let's do the boost in the x-direction

$$x'' = \gamma_v(x' - vt'),\ ct'' = \gamma_v(ct' - \beta_v x'),\ z'' = z', y'' = y'$$

Now let's substitute to get the form in question, because $x' = x$ and $t' = \gamma_u(t - \frac{uz}{c^2})$.
We get for x''

$$x'' = \gamma_v \left[x - v\gamma_u \left(t - \frac{uz}{c^2}\right)\right] = \gamma_v[x - \beta_v\gamma_u(ct - \beta_u z)]$$

now for z''

$$z'' = \gamma_u(z - ut)$$

for y''

$$y'' = y' = y$$

lastly for ct''

$$ct'' = \gamma_v[\gamma_u(ct - \beta_u z) - \beta_v x] = \gamma_v\gamma_u \left[ct - \left(\beta_u z + \frac{\beta_v x}{\gamma_u}\right)\right]$$

b)
Let's show that $\gamma_u\gamma_v = \gamma_w$. From the given definition we have

$$\mathbf{w} = \mathbf{u} + \frac{1}{\gamma_u}\mathbf{v}$$

Thus,

$$w^2 = u^2 + \frac{v^2}{\gamma_u^2}$$

because $\mathbf{u}$ and $\mathbf{v}$ are orthogonal with each other. Notice that because

$$\gamma_u^2 = \frac{1}{1 - \frac{u^2}{c^2}}$$

then

$$\frac{v^2}{\gamma_u^2} = v^2\left(1 - \frac{u^2}{c^2}\right)$$

Thus,

$$\frac{w^2}{c^2} = \frac{u^2}{c^2} + \frac{v^2}{c^2} - \frac{u^2v^2}{c^4}$$

$$1 - \frac{w^2}{c^2} = \left(1 - \frac{u^2}{c^2}\right)\left(1 - \frac{v^2}{c^2}\right)$$

Take the square root, and we get

$$\gamma_w = \frac{1}{\sqrt{1 - \frac{w^2}{c^2}}} = \gamma_u\gamma_v$$

c)
If this is a pure boost in the $\hat{w}$-direction, then it must be true that

$$c_w z'' + s_w x'' = \gamma_w(c_w z + s_w x - wt)$$

with

$$\hat{w} = c_w\hat{z} + s_w\hat{x}$$

Now let's check the coefficient of z.
Let's calculate the left hand side, focusing only on the coefficients of x.

$$c_w z'' + s_w x'' = c_w\gamma_u(z - ut) + s_w\gamma_v[x - \beta_v\gamma_u(ct - \beta_u v)]$$

Left hand side, we have

$$s_w\gamma_v x$$

Now let's observe the right hand side, the z coefficient must be

$$\gamma_w s_w x$$

Now let's compare, for them to be equal this equation must hold true

$$\gamma_v s_w = \gamma_w s_w \Rightarrow \gamma_v = \gamma_w$$

$$\gamma_v = \gamma_v\gamma_u$$

$$\gamma_u = \frac{1}{\sqrt{1 - \frac{u^2}{c^2}}} = 1$$

However, this is never true. The reason is that u is non-zero, or $\gamma_u \neq 1$
d)
If we instead flip the order of the boost, say doing x-direction and then the z-direction. We have for the $\mathbf{w}$

$$\mathbf{w}^\star = \frac{\mathbf{u}}{\gamma_v} + \mathbf{v}$$

It is obvious that $\mathbf{w}^\star \neq \mathbf{w}$ as the gamma factor appears in different components. However, the magnitude

$$|\mathbf{w}^\star|^2 = \frac{u^2}{\gamma_v^2} + v^2$$

is the same as

$$|\mathbf{w}^\star|^2 = u^2\left(1 - \frac{v^2}{c^2}\right) + v^2 = u^2 + v^2 - \frac{u^2v^2}{c^2}$$

$$|\mathbf{w}|^2 = u^2 + v^2 \left(1 - \frac{u^2}{c^2}\right) = u^2 + v^2 - \frac{u^2 v^2}{c^2}$$